Letting Time Lie Down: Visualizing and Quantifying Simple Harmonic Motion

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The sinusoidal displacement–time graph of simple harmonic motion (SHM) is difficult to show directly in the laboratory. We designed a three-step approach that turns time into a horizontal distance. First, a laser pointer attached to a spring–mass oscillator writes a glowing sinusoid on a phosphorescent foil that a small motor tows past it. Second, a free browser tool, the *Frame Sequencer*, crops the region of motion from successive frames of a smartphone video and tiles these strips side by side, so that the sinusoid appears directly and its period can be read off. Third, in a quantitative classroom activity, students filmed masses of 150–400 g hanging from one spring and from two springs in series. The period squared increases linearly with mass, and the ratio of the two slopes, 1.99 ± 0.06, agrees with the predicted value of 2.

The problem of "seeing" SHM

Simple harmonic motion sits at the heart of the introductory physics curriculum, and students meet it again whenever they study waves, sound, or AC circuits. Unlike projectile motion, whose parabolic trajectory already gives students a two-dimensional picture, one-dimensional oscillation offers only a back-and-forth along a single line, and the canonical sinusoidal y–t graph is not actually visible in the laboratory. McDermott, Rosenquist, and van Zee documented long ago that translating between motion and graphs is a persistent student difficulty,[1] subsequently confirmed by Beichner.[2] Existing remedies include low-cost mechanical tracers,[3] Arduino-based distance sensors,[4] smartphone accelerometers,[5] video-analysis packages such as Tracker,[6] and phosphorescent-foil displays for Lissajous figures.[7]

Step 1 — See it: a glowing sinusoid on the bench

The first step is a sensory demonstration: a simple analog apparatus that lets students watch a sinusoid being drawn in real time (Fig. 1).

Apparatus. Two identical springs are suspended in parallel from a retort stand, their lower ends joined by a 3D-printed bracket, or more cheaply a wooden block drilled to the laser's diameter. The bracket holds a small violet laser pointer (405 nm, <1 mW, Class 2, 21 CFR compliant) with its beam horizontal, and the hook masses hang from the bracket below it; the parallel arrangement keeps the bracket from rotating about a vertical axis as it oscillates. As with any Class 2 source, students are instructed not to look directly into the beam.

Physical time axis. A phosphorescent foil of roughly letter-paper size (about 20 cm × 30 cm) is mounted vertically on a low-friction cart towed along the bench by a thread wound onto the output shaft of a small DC gear motor; the reduction gearbox gives a slow, steady pull of about 11 cm/s, constant to within about 3% as measured by video analysis. The hook masses are chosen so that the oscillation period is about 1 s, convenient for observation at typical cart speeds.

Observations. As the cart moves horizontally (the time axis) and the oscillator moves vertically (the displacement axis), the laser briefly excites the phosphor and leaves a glowing afterimage that traces out a clean sinusoid (Fig. 1). The trace persists for nearly a minute. Students see the sinusoidal shape emerge in front of them before any equation is written or any number measured. A short video of the recorder in operation is provided as Supplementary Material.

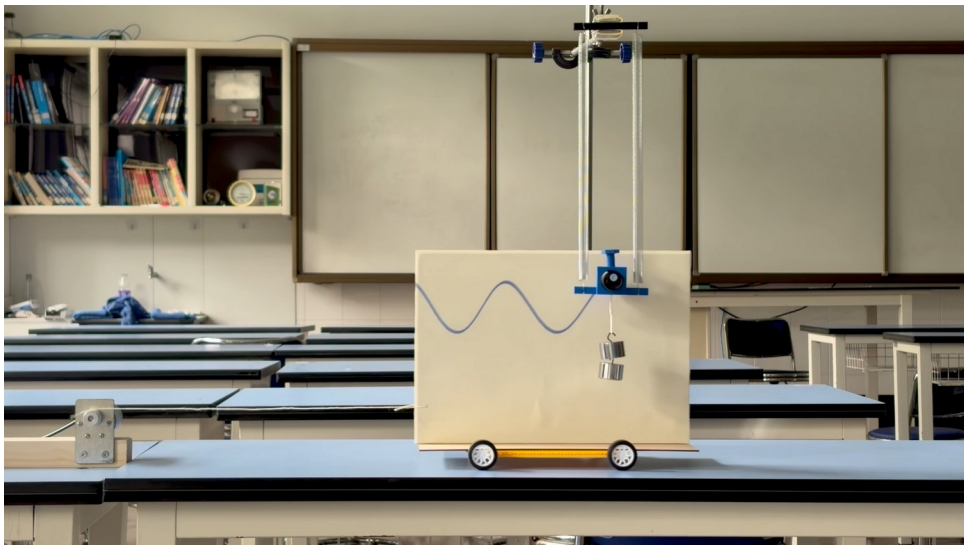


Fig. 1. The laser–phosphor recorder during a run. A violet laser pointer below a parallel-spring oscillator writes a glowing sinusoid on the phosphorescent foil mounted on a cart towed along the bench at about 11 cm/s.

Step 2 — Slice it: every phone is an oscilloscope

Once the connection between SHM and the sinusoid has been seen, students need a tool to capture the same motion quantitatively. We built the *Frame Sequencer* (Fig. 2), a free, browser-based tool that does in software what the cart does in hardware, so students who have watched the cart lay out the time axis can see how its images are built. The live version is hosted at https://frame-sequencer-909934852596.us-west1.run.app and is actively maintained by the authors; a stable offline build that will not be updated further, a screen recording of the workflow, and the sample spring–mass video used in this section are provided as Supplementary Material.

Workflow. In practice, the user uploads a short smartphone clip; 5 to 10 seconds is usually enough. To minimize camera shake from pressing the record button, the tool defaults to a 3-second window centered within the clip, sampled at 15 fps; the window length, position, and sampling rate are user-adjustable. The tool compares successive frames to locate the region of

motion automatically (purple box in the multi-frame overlay at the top of Fig. 2), with manual selection also supported; the phone should therefore be kept still, the lighting steady, and nothing else, such as hands, people, or shadows, should move in the frame. Each sampled frame is cropped to that region, and the cropped strips are tiled side by side in chronological order. Because every strip has the same width and the frames are sampled at equal time intervals, the horizontal coordinate maps linearly to physical time t via the sampling rate, and the vertical axis corresponds to position, so the composite is a true displacement–time image $y(t)$ rather than a displacement–pixel plot. The tool then locates the object in each strip by scanning upward from the bottom for its lower edge (red dots in Fig. 2), so a plain background that contrasts clearly with the object gives the most reliable results. Finally, it overlays a best-fit curve (green) of the chosen model (linear, quadratic, or sinusoidal), returning the equation and R^2.

Observations. The bottom panel of Fig. 2 is the composite produced from a sample spring–mass video. The periodic pattern is obvious by eye, and the sinusoidal fit returns

$$y(t) = (229.31\,\text{px})\sin[(8.1263\,\text{rad/s})\,t + 3.72\,\text{rad}] + 1200.33\,\text{px}, \tag{1}$$

with $R^2 = 0.9967$, corresponding to a period $T = 2\pi/\omega \approx 0.77$ s; all values are as returned by the tool. The phase φ and offset c, absent from the textbook form $A\sin\omega t$, arise because the clip starts at an arbitrary moment and y is measured in pixels downward from the top of the composite image; $c = 1200.33$ px thus marks the equilibrium position of the tracked lower edge.

Analysis by hand. Detection and fitting are optional steps applied after the composite is built; we used them so that the Step 3 activity fit into about 20 min of class time. When more time is available, students can print the composite and analyze it themselves, much as with a strobe photograph: because successive strips are 1/15 s apart at the default sampling rate, the

period follows from counting the strips spanning several cycles, and the automatic fit then serves as a check.

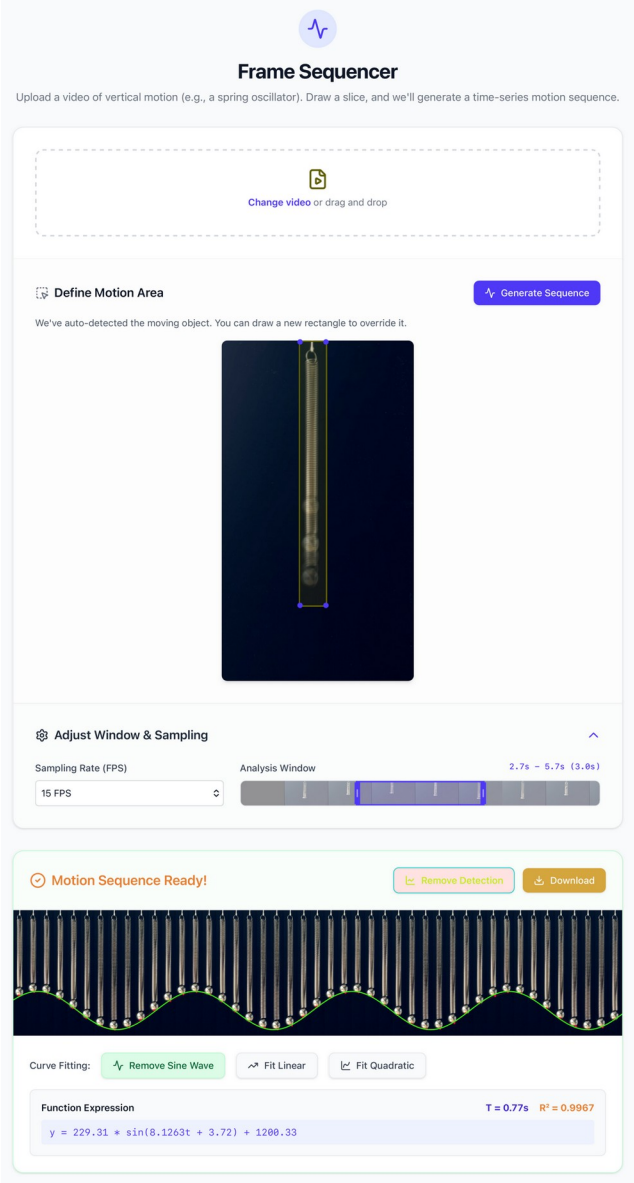


Fig. 2. The Frame Sequencer workflow. Top: a multi-frame overlay of the uploaded video, with the moving object detected automatically (purple box). Middle: a 3-s window sampled at 15 fps. Bottom: cropped strips tiled in time order, the object tracked across strips (red dots), and a sinusoidal fit (green) returning $y(t) = A\sin(\omega t + \varphi) + c$ with T and R^2; in the displayed equation, y is in pixels and t in seconds.

Step 3 — Solve it: the class as collaborators

With a tool that returns a numerical period from any video, it becomes practical for a whole class to test the SHM prediction quantitatively, working as a team. Students in one author's class recorded smartphone videos under the instructor's guidance, in two spring configurations: a single spring (supplier's value $k = 24$ N/m, confirmed beforehand by the instructor in a static Hooke's-law test), and two identical springs in series ($k_{\text{eff}} = k/2$). The suspended mass was varied from 150 g to 400 g in 50-g steps. For each mass-and-spring combination the video was uploaded to the Frame Sequencer and the period T read off from the automatic sinusoidal fit; most fits returned $R^2 > 0.99$. Working in groups, students were given about 15 min of class time to record their videos and obtain the periods; the instructor then pooled the results in a spreadsheet and plotted T^2 against m in front of the class in about 5 min, so the whole activity took about 20 min. Fig. 3 shows four of the twelve outputs, for the lightest (150 g) and heaviest (400 g) masses: for each spring configuration the period T grows as m increases, and halving the effective stiffness (upper pair to lower pair) lengthens T by a factor of $\sqrt{2}$.

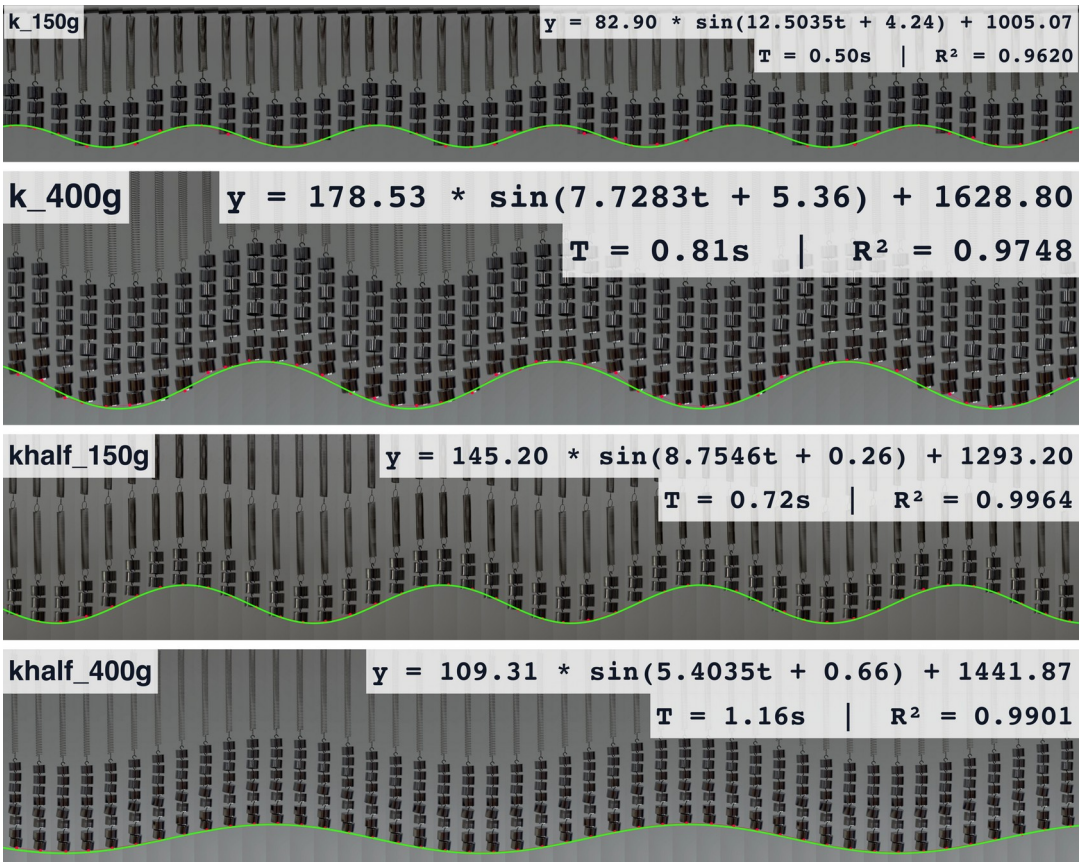


Fig. 3. Frame Sequencer outputs for four of the twelve student videos: the lightest (150 g) and heaviest (400 g) masses on a single spring (labeled k, upper pair) and on two springs in series (labeled khalf, lower pair). Each panel spans the same 3-s window and shows the time-stacked composite with the detected position (red) and the best-fit sinusoid (green). Labels are as exported by the tool: configuration and mass at upper left; fitted equation (y in pixels, t in seconds), period T, and R^2 at upper right. All twelve outputs are provided as Supplementary Material.

Linear model. For an oscillator of mass m on springs of effective stiffness k_{eff}, simple harmonic motion predicts

$$T^2 = \frac{4\pi^2}{k_{\text{eff}}} m, \tag{2}$$

so T^2 versus m should fall on a straight line with slope $4\pi^2/k_{\text{eff}}$. The pooled student data are plotted in Fig. 4.

Linearity. Both configurations sit on clean straight lines, with $R^2 = 0.9974$ (single spring) and $R^2 = 0.9989$ (series pair). The high R^2 values confirm the predicted $T^2 \propto m$ relationship across the full mass range.

Slope ratio. Going from a single spring ($k_{\text{eff}} = k$) to two identical springs in series ($k_{\text{eff}} = k/2$) should multiply the slope by exactly 2.00. The measured slope ratio is 1.99 ± 0.06, matching the theoretical value to within 1%.

Non-zero intercept and the spring's own weight. Both fitted lines show small positive intercepts rather than passing exactly through the origin. This is consistent with the mass of the spring itself: a spring of mass m_s contributes an effective mass $m_s/3$ to the oscillator,[8] so Eq. (2) generalizes to

$$T^2 = \frac{4\pi^2}{k_{\text{eff}}}\left(m + \frac{m_{s,\text{eff}}}{3}\right), \tag{3}$$

shifting the line upward by $(4\pi^2/k_{\text{eff}})(m_{s,\text{eff}}/3)$ at $m = 0$. The series configuration, with halved k_{eff} and twice the active spring material, has the larger observed intercept, in qualitative agreement with Eq. (3).

Consistency check — recovering k. Inverting the fitted slopes through $k_{\text{eff}} = 4\pi^2/\text{slope}$ provides an independent measurement of the spring constant from each configuration. The single-spring slope of 1.65 s²/kg gives $k \approx 23.9$ N/m; the series-pair slope of 3.28 s²/kg gives $k_{\text{eff}} \approx 12.0$ N/m, hence $k \approx 24.1$ N/m. The two values agree with each other and with the supplier's value $k = 24$ N/m to within $\sim 0.5\%$, providing an internally consistent verification of the SHM model.

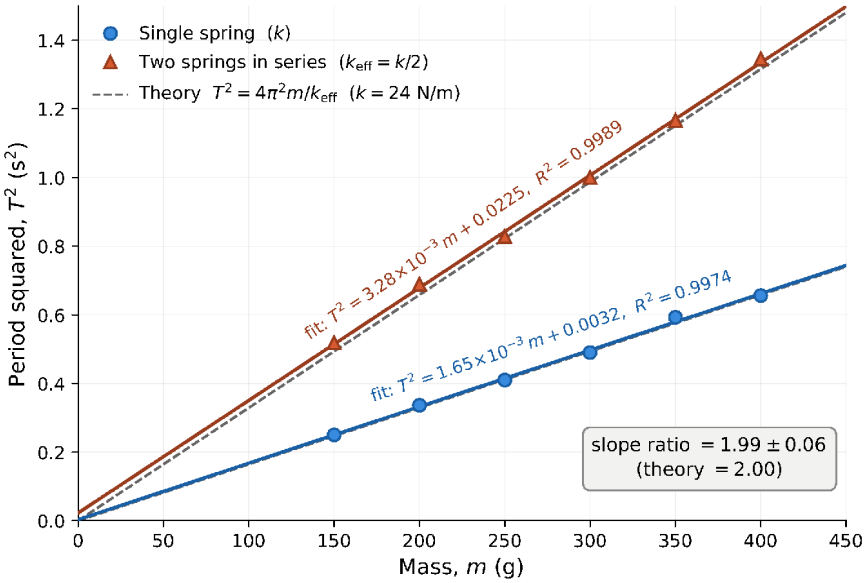


Fig. 4. Period squared T^2 versus suspended mass m for the single-spring (blue circles) and two-springs-in-series (orange triangles) configurations. Solid lines are linear fits to the data ($R^2 = 0.9974$ and 0.9989); dashed gray lines are the parameter-free SHM predictions $T^2 = 4\pi^2 m / k_{\text{eff}}$ obtained by substituting the supplier's value $k = 24$ N/m into Eq. (2), not fitted to the data.

Supplementary Material

Readers can access the supplementary material at TPT Online. It contains:

- analog_demo.mp4 — the laser–phosphor recorder of Step 1 in operation.
- frame-sequencer.html — an offline build of the Frame Sequencer, frozen as a stable snapshot.
- frame_sequencer_workflow.mp4 — a short screen recording of the Frame Sequencer workflow.
- spring_mass_sample.mp4 — the sample video of Fig. 2.
- all_student_results.pdf — the outputs for all twelve student videos of Step 3.
- phi_c_animation.mp4 — an animation explaining the phase φ and offset c in Eq. (1).

- gray_wall_sample.mp4 — a student video recorded against a plain classroom wall, for trying the tool under ordinary conditions.

References

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